# Thyroid storm — Q83

## How is thyroid storm diagnosed

- Clinical diagnosis: thyrotoxicosis with organ decompensation (hyperthermia, CNS dysfunction, tachycardia/AF/heart failure, GI–hepatic dysfunction)
- Burch–Wartofsky score ≥ 45 highly suggestive (25–44 impending)
- Precipitant: surgery, infection, childbirth, iodine contrast, stopping antithyroid drugs

## Outline the management in order, with drugs and doses

- β-blocker: propranolol 60–80 mg orally 4–6-hourly (also blocks T4→T3) or esmolol infusion if heart failure concern
- Thionamide: PTU 500–1000 mg load then 250 mg 4-hourly (blocks synthesis and peripheral conversion) or methimazole 20 mg 4–6-hourly
- Iodine (Lugol's or potassium iodide) at least 1 h AFTER the thionamide – blocks release of stored hormone
- Hydrocortisone 100 mg 8-hourly (reduces T4→T3; treats relative adrenal insufficiency)
- Supportive: cooling, paracetamol (avoid aspirin – displaces T4 from binding proteins), fluids, treat precipitant; cholestyramine; plasmapheresis/thyroidectomy if refractory

## Why is iodine given 1 hour after the thionamide

- Iodine given first provides substrate for new hormone synthesis → can worsen thyrotoxicosis
- Thionamide first blocks thyroid peroxidase (organification), so the iodine cannot be used to make more hormone
- Iodine then inhibits release of preformed hormone (Wolff–Chaikoff effect)

# Thyroid storm — Q84

## Explain the mechanisms of thyrotoxic heart failure

- High-output state: increased metabolic demand, reduced SVR, increased blood volume
- Tachycardia-mediated cardiomyopathy (AF with rapid rate) → low-output failure
- Direct T3 effects on myocyte genes (α-myosin, SERCA) and arrhythmias

## How would you manage his rate and heart failure

- Short-acting β1 blocker (esmolol) titrated – rapid off if hypotension
- Avoid large doses of propranolol in decompensated heart failure; consider digoxin (higher doses needed) or amiodarone (caution: iodine load)
- Diuretics and ventilatory support as needed
- Anticoagulation for AF (high stroke risk in thyrotoxicosis)

## When are second-line therapies indicated

- Failure of medical therapy in 24–48 h or intolerance of thionamides (agranulocytosis, hepatotoxicity)
- Therapeutic plasma exchange to remove circulating hormone
- Urgent thyroidectomy after preparation; ECMO for cardiogenic shock

# Myxoedema coma — Q85

## What are the precipitants and clinical features of myxoedema coma

- Precipitants: infection, cold, surgery, stopping thyroxine, sedatives/opioids, stroke, amiodarone, lithium
- Features: decreased consciousness, hypothermia, bradycardia, hypotension, hypoventilation
- Hyponatraemia, hypoglycaemia, pericardial effusion, ileus, non-pitting oedema

## What is the first drug treatment? Explain the order

- Hydrocortisone 100 mg IV 8-hourly FIRST (or together) until adrenal insufficiency excluded
- Thyroxine increases cortisol metabolism → may precipitate adrenal crisis
- IV levothyroxine 200–400 µg loading dose (lower in elderly/cardiac), then 1.6 µg/kg/day (75% IV)
- ± IV liothyronine (T3) 5–20 µg then 2.5–10 µg 8-hourly in selected patients (arrhythmia risk)

## Outline the supportive management

- Intubation/ventilation for hypoventilation and hypercapnia
- Passive rewarming (active rewarming can cause vasodilatation and collapse)
- Treat hyponatraemia (fluid restriction; hypertonic saline if severe/symptomatic) and hypoglycaemia
- Look for and treat infection; avoid sedatives

# Myxoedema coma — Q86

## Explain how amiodarone affects thyroid function

- Contains 37% iodine by weight → large iodine load
- Hypothyroidism: Wolff–Chaikoff effect (failure to escape), especially with autoimmune thyroiditis
- Thyrotoxicosis: type 1 (iodine-induced synthesis) and type 2 (destructive thyroiditis)

## Why does hypothyroidism cause failure to wean

- Reduced hypoxic and hypercapnic ventilatory drive
- Respiratory muscle weakness (myopathy), diaphragm dysfunction
- Macroglossia, upper airway oedema, pleural effusions, obesity; sleep apnoea

## How would you treat his hypothyroidism safely

- Hydrocortisone cover until cortisol checked
- Cautious levothyroxine replacement (lower starting dose in elderly/cardiac disease) – risk of ischaemia and arrhythmia
- Review need for amiodarone (cardiology)
- Drain the effusion if causing tamponade; monitor ECG

# DKA with hypokalaemia — Q87

## State the diagnostic criteria for DKA

- Ketonaemia ≥ 3 mmol/L (or ketonuria ≥ 2+)
- Glucose > 11 mmol/L (200 mg/dL) or known diabetes
- pH < 7.3 and/or HCO3 < 15 mmol/L
- She has severe DKA (pH < 7.1)

## What is the next step after starting IV fluids? Explain why

- Give IV potassium replacement and DELAY insulin until K ≥ 3.3–3.5 mmol/L
- Insulin drives K into cells → fatal hypokalaemia (arrhythmia, respiratory muscle weakness)
- Potassium 20–40 mmol/h (via central line if high concentration), with cardiac monitoring
- Once K ≥ 3.5: fixed-rate insulin 0.1 U/kg/h; continue K in fluids (40 mmol/L when K 3.5–5.5)

## Describe the fluid regimen, targets and resolution criteria

- 0.9% saline or balanced crystalloid 1 L over 1 h, then 1 L over 2 h, 2 h, 4 h, 4 h (adjust to age/heart)
- Add 10% dextrose when glucose < 14 mmol/L, continuing insulin to clear ketones
- Targets: ketones fall ≥ 0.5 mmol/L/h, HCO3 rise ≥ 3 mmol/L/h, glucose fall 3 mmol/L/h
- Resolution: ketones < 0.6 mmol/L and pH > 7.3 (HCO3 > 15)
- Bicarbonate not recommended unless pH < 6.9–7.0

# DKA with hypokalaemia — Q88

## What is the diagnosis and why does it happen

- Euglycaemic DKA associated with SGLT2 inhibitors
- SGLT2 inhibitors cause glycosuria (lower glucose), reduce insulin and raise glucagon → ketogenesis
- Precipitated by surgery, fasting, acute illness, dose reduction of insulin

## How does management differ from classic DKA

- Stop the SGLT2 inhibitor
- Start dextrose (10%) early with the fixed-rate insulin because glucose is not high
- Diagnosis is missed if relying on glucose – check ketones in ill patients on SGLT2i

## Compare DKA and HHS

- HHS: glucose > 30 mmol/L (540 mg/dL), osmolality > 320 mOsm/kg, no significant ketosis (< 3 mmol/L), pH > 7.3
- HHS: profound dehydration (100–220 mL/kg), older patients, higher mortality
- HHS: fluid first; insulin only when glucose stops falling with fluids (0.05 U/kg/h)
- HHS: aim slow correction (osmolality fall 3–8 mOsm/kg/h, Na fall ≤ 10 mmol/L/24 h); high VTE risk – prophylactic LMWH

# Candidaemia — Q89

## Outline the treatment of candidaemia

- Echinocandin (caspofungin, micafungin, anidulafungin) as initial therapy
- Step down to fluconazole after 5–7 days if isolate susceptible and patient clinically stable with negative repeat cultures
- Remove the central venous catheter as soon as possible
- Treat for 14 days after the first negative blood culture and resolution of symptoms

## What further investigations are needed

- Repeat blood cultures daily or every other day until negative
- Dilated fundoscopy (ophthalmology) within the first week to exclude endophthalmitis
- Echocardiography (TTE/TOE) if persistent candidaemia or suspicion of endocarditis

## List the risk factors for invasive candidiasis in the ICU

- Central venous catheters, total parenteral nutrition
- Broad-spectrum antibiotics, abdominal surgery (especially anastomotic leak), pancreatitis
- Candida colonisation at multiple sites, renal replacement therapy
- Immunosuppression, steroids, prolonged ICU stay

# Candidaemia — Q90

## How would you decide whether to start empirical antifungal therapy

- Candida score (TPN 1, surgery 1, multifocal colonisation 1, severe sepsis 2) ≥ 3 → high risk
- Colonisation index ≥ 0.5 (number of sites colonised / sites cultured)
- Serum β-D-glucan (high negative predictive value) to support or rule out
- Empirical echinocandin reasonable in high-risk septic patients; stop if β-D-glucan negative and no improvement (EMPIRICUS: no benefit in unselected patients)

## What are the limitations of blood cultures and β-D-glucan

- Blood cultures positive in only ~50% of invasive candidiasis
- β-D-glucan false positives: haemodialysis cellulose membranes, albumin/IVIG, gauze, bacteraemia
- β-D-glucan not specific to Candida (also Aspergillus, Pneumocystis)

## Name the Candida species with resistance concerns

- C. glabrata (Nakaseomyces): reduced fluconazole susceptibility, emerging echinocandin resistance
- C. krusei (Pichia kudriavzevii): intrinsically fluconazole resistant
- C. auris: multidrug resistant, outbreaks, needs infection control

# Toxic shock syndrome — Q91

## What is the diagnosis and the pathophysiology

- Staphylococcal toxic shock syndrome (menstrual)
- Exotoxin TSST-1 (or enterotoxins) acts as a superantigen
- Binds MHC class II and T-cell receptor outside the antigen site → massive T-cell activation and cytokine storm → shock and multi-organ failure

## Outline the antibiotic and adjunctive treatment, explaining the role of clindamycin

- Remove source (tampon), drain/debride any focus
- β-lactam with anti-staphylococcal activity (flucloxacillin/cefazolin; vancomycin if MRSA risk)
- PLUS clindamycin 600–900 mg IV 8-hourly
- Clindamycin inhibits protein synthesis (50S) → stops toxin production; effective regardless of inoculum size (no Eagle effect)
- IVIG (neutralising antibodies against superantigens) in severe/refractory cases, especially streptococcal TSS

## Compare staphylococcal and streptococcal TSS

- Strep TSS: usually from invasive soft-tissue infection (necrotising fasciitis), severe pain
- Strep: blood cultures positive in > 50%; staph: usually negative
- Strep TSS mortality higher (30–70%)
- Staph: desquamation of palms/soles 1–2 weeks later

# Toxic shock syndrome — Q92

## What is the likely diagnosis and how would you confirm it

- Necrotising soft tissue infection (necrotising fasciitis), possibly with streptococcal toxic shock
- Clinical diagnosis: pain out of proportion, rapid spread, bullae, crepitus, systemic toxicity
- LRINEC score ≥ 6 suggestive (poor sensitivity); CT may show gas; do not delay surgery for imaging

## Describe the management

- Urgent surgical debridement (within hours) and repeated re-look surgery
- Broad-spectrum antibiotics: piperacillin–tazobactam or meropenem + clindamycin (+ vancomycin/linezolid if MRSA)
- Resuscitation and organ support
- IVIG in streptococcal TSS (some evidence); hyperbaric O2 not routinely recommended

## Classify necrotising fasciitis by microbiology

- Type I: polymicrobial (anaerobes + Gram-negatives), diabetics, perineal (Fournier's)
- Type II: monomicrobial – group A Streptococcus (± S. aureus)
- Type III: Clostridium (gas gangrene) or Vibrio vulnificus